MIL-STD-1316C
3 January 1984
SUPERSEDING
MIL-STD-1316B
15 February 1977

MILITARY STANDARD

FUZE DESIGN,
SAFETY CRITERIA FOR



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

FSC 1390

MIL-STD-1316C
3 January 1984

DEPARTMENT OF DEFENSE

Washington, D.C. 20301

Fuze Design, Safety Criteria for

MIL-STD-1316C

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, Armament Research and Development Center, US Army AMCCOM, ATTN: DRSMC-TST-S(D), Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. Comments should be forwarded through the designated Reviewing Activity listed for each Service in 7.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

FOREWORD

The purpose of this standard is to establish specific design safety criteria for fuzes. It applies primarily to the safing and arming functions performed by fuzes intended for use with munitions. The safety and arming requirements and objectives specified herein are mandatory fundamental elements of design, engineering, production and procurement of fuzes. Fuzes shall provide safety which is consistent with assembly, handling, storage, transportation, operational readiness and use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

MILITARY STANDARD

FUZE DESIGN,
SAFETY CRITERIA FOR

TO ALL HOLDERS OF MIL-STD-1316C:

1. The following pages of MIL-STD-1316C have been revised and supersede the pages listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
i	28 September 1987	i	3 January 1984
ii	28 September 1987	ii	3 January 1984
7	28 September 1987	7	3 January 1984
8	28 September 1987	8	3 January 1984
9	28 September 1987	9	3 January 1984
10	28 September 1987	10	3 January 1984
10A	28 September 1987	-	-
11	28 September 1987	11	3 January 1984
12	28 September 1987	12	3 January 1984

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-1316C will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appendix pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or cancelled.

Custodians:
Army - AR
Navy - OS
Air Force - 11

Review Activities:
Army - MI
Navy - AS
Air Force - 18, 99

Preparing Activity:
Army - AR

(Project 1390-0553)

AMSC N/A

FSC 1390

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

MIL-STD-1316C
NOTICE 1

DEPARTMENT OF DEFENCE
Washington, D.C. 20301

Fuze Design, Safety Criteria for

MIL-STD-1316C

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Picatinny Arsenal, NJ 07806-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. Comments should be forwarded through the designated Reviewing Activity listed for each Service in 7.2.

Supersedes page i of MIL-STD-1316C

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

FOREWORD

The purpose of this standard is to establish specific design safety criteria for fuzes. It applies primarily to the safing and arming functions performed by fuzes intended for use with munitions. The safety and arming requirements and objectives specified herein are mandatory fundamental elements of design, engineering, production and procurment of fuzes. Fuzes shall provide safety which is consistent with assembly, handling, storage, transportation, operational readiness and use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

3.1.32 Stored energy. A component contains stored energy if the component is capable of delivering energy in addition to the external energy required to initiate its function. Examples of stored energy components are:

a. Batteries

b. Charged firing capacitors

c. Compressed gas

d. Explosive actuators (bellows and dimple motors, etc.)

e. Loaded springs

4. REQUIREMENTS

4.1 General. The features, procedures or controls listed in this section shall be used in the design of fuzes included in the scope of this document.

4.2 Fuze safety system.

4.2.1 Safety redundancy. The safety system of fuzes shall contain at least two independent safaety features, each of which shall prevent unintentional arming of the fuze. The force enabling a minimum of two safety features shall be derived from different environments. An action taken to initiate launch may be considered an environment if the signal generated by the action irreversibly commits the ammunition to complete the launch cycle. Operation of at least one of these safety features shall depend on sensing an environment after first motion in the launch cycle or on sensing a post-launch environment.

4.2.2 Arming delay. A safety feature of the fuze shall provide an arming delay. Arming delay when used as the criterion for safety shall assure that a safe separation distance can be achieved for all defined operational conditions.

4.2.3 Manual arming. An assembled fuze shall not be capable of being armed manually.

4.3 Explosive trains.

4.3.1 Compatibility of fuze elements. Explosive compositions in fuzes shall be qualified for use in accordance with ADA-086259, Vol. 4 in their intended roles in interrupted or non-interrupted explosive train components. All fuze materials shall be chosen to be compatible and stable so that under all life-cycle conditions none of the following can occur in an unarmed fuze:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

a. Premature arming

b. Dangerous ejection of material

c. Deflagration or detonation of the lead or booster

d. An increase in the sensitivity of explosive train components beyond the level at which they were approved for service use and at which they are customarily used.

e. Compromise of safety or sterilization features

4.3.2 Explosive sensitivity (lead and booster explosives). Only explosives subjected to the tests of ADA-086259, Vol. 4, Chapters 3 and 4 and specifically approved by the associated safety evaluation board of 7.2 for each application are permitted in a position leading to the initiation of a high explosive main charge without interruption, when the fuze is in the safe condition. The explosive material used in fuze systems shall not be altered by any means (precipitation, recrystallization, grinding, density changes, etc.) likely to increase its sensitivity beyond that at which the material was approved for service use and at which it is customarily used. Table 1 contains a list of currently approved explosives for all Services.

TABLE. Approved Explosives for all Services

Explosive	Specification
Comp A3	MIL-C-440
Comp A4	MIL-C-440
Comp A5	MIL-E-14970
Comp CH6	MIL-C-21723
PBXN-5	MIL-E-81111
PBXN-6	WS-12604
DIPAM	WS-4660
HNS Type 1 or Type 2 GR A	WS-5003
*Tetryl	MIL-T-339
*Tetryl Pellets	MIL-P-46464

*No longer manufactured. Not for use in new developments.

4.3.3 Explosive train interruption. When the explosive train contains explosive material other than those allowed by 4.3.1, at least one interrupter (shutter, slider, rotor) shall separate those explosives more sensitive than those allowed by 4.3.1 from the lead and booster until the arming sequence is completed. The interrupter(s) shall be directly locked mechanically in the safe position by at least two independent safety features. The safety features shall not be removed until the arming sequence begins. If the initiating explosive material is housed in the interrupter, a single interrupter locked by the two independent safety features is acceptable. If the sensitive element is positioned such that safety is completely dependent

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

upon the presence of an interrupter, the design shall include positive means to prevent the fuze from being assembled without the properly positioned interrupter. The effectiveness of interruption shall be determined by techniques described in MIL-STD-331, Test 115, including Section 6 or equivalent to Section 6, during design evaluation.

4.3.4 Non-interrupted explosive train control. When the explosive train contains only those explosive materials allowed by 4.3.1, no explosive train interruption is required. One of the following methods of controlling function energy shall be employed to preclude arming before safe separation. Additionally, the fuze design shall include positive means to prevent the fuze from being assembled without its energy control feature(s). The combined probability of having minimum function energy in the fuze, having a failure of the energy control feature(s) and firing the initiator with minimum function energy must be compatible with the specified fuze safety system failure rate (see 4.6).

a. For fuzes containing minimum non-electrical function energy prior to safe separation, at least one energy interrupter directly and mechanically locked in the safe position by at least two independent safety features shall prevent the flow of energy to the initiator.

b. For fuzes containing minimum electrical function energy prior to safe separation, at least two energy interrupters directly and mechanically locked in the safe position, each by an independent safety feature, shall prevent the flow of electrical energy to the initiator.

c. For systems using techniques for accumulating functioning energy from the post-launch environment, the fuze shall not permit any functioning energy to reach the initiator until verification, by the fuze, of a proper launch, and attainment of a safe separation distance. Additionally, any energy of the type required to function the initiator which exists in the fuze prior to safe separation shall be less than the minimum function energy.

4.3.5 Electrical sensitivity. The initiator for an electrically fired non-interrupted explosive train shall meet the characteristics listed for Class B initiators of MIL-I-23659. The initiator shall not be capable of being initiated by any electrical signal at a potential of less than 500 volts. Electromagnetic emission sensitivity and susceptibility of the fuze shall not create a hazard. The requirements of MIL-STD-461 (and DOD-STD-1463 for the Army) shall apply.

4.4 Logic functions. Any logic functions related to safety elements performed by the fuze shall be embedded as firmware or hardware.

4.5 Safe or armed condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

4.5.1 Safe-arm options. In order to allow the safe condition of a fuze to be assured, fuzing system designs shall incorporate one or more of the following features:

a. Prevention of the assembly of the fuzing system in other than a safe condition.

b. Assurance of a positive means of determining the safe condition of the system during and after assembly and when installing the system into a munition.

c. Prevention of the installation of an armed, assembled fuzing system into a munition.

If arming and reset of the assembled fuzing system in tests is a normal procedure in manufacturing, inspection, or at any time prior to its installation into a munition, prevention of unsafe assembly (subparagraph a) is not sufficient and the provisions of either subparagraph b or c must also be met.

4.5.2 Visual indication. If visual indication of the safe-armed condition is to be employed in the fuze, visible indicators shall be color coded (see 7.1) to show the fuze condition as follows:

a. Safe condition. Fluorescent green background with the letter S or word SAFE superimposed thereon in white. Colors shall be nonspecular.

b. Armed condition. Fluorescent red background with the letter A or the word ARMED superimposed thereon in black. Colors shall be nonspecular.

4.6 Safety system failure rate. The safety system failure rate of a fuze shall be calculated by performing a safety analysis (see 6.3) and shall be verified to the extent practicable by test and analysis during fuze evaluation. As a minimum requirement, the safety system failure rate shall not exceed one failure in one million prior to intentional initiation of the arming sequence. A minimum quantitative requirement is not specified for the time frame after launch to the attainment of the proper arming delay since it will be dependent upon the particular weapon and its use. Fuze malfunctions which occur after the proper arming delay shall not be included in the fuze safety system failure rate.

4.7 Documentation. The evaluation program used as the basis of the safety assessment which is prepared by the developing agency shall be documented in both detail and summary form.

4.8 Design for quality control and inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

4.8.1 Fuzes shall be designed and documented to facilitate application of effective quality control and inspection procedures. Design characteristics critical to fuze safety shall be identified to assure that the designed safety is maintained.

4.8.2 The design of the fuze shall facilitate the use of inspection and test equipment for all critical design characteristics identified.

4.9 Design approval. At the inception of engineering development, the weapon system program manager shall obtain approval of the cognizant service authority for weapon safety (and other service authorities, for multi-service application) of both the design concept and the methodology for assuring a comprehensive safety assessment. At the completion of engineering development, the adequacy of the safety assessment shall require the review and approval of the cognizant service weapon authority(ies), (see 7.2).

5. OBJECTIVES

5.1 General. The features, procedures, or controls listed as objectives in this section are to be accepted as requirements in the design of fuzes covered by the scope of this document when achievement is within the sate of the art of fuze development and is feasible within the scope of the development. Necessity for deviations from these objectives shall be documented, showing that compliance is not feasible (see 7.2).

5.2 Design features.

5.2.1 Stored energy. The fuze safety system should not utilize stored energy to remove safety features or provide arming energy unless no adequate environmentally derived energy is available. If stored energy is used, it must be demonstrated that safety is not compromised.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

5.2.2 Fail-safe features. Fuze systems should incorporate fail-safe features based onltheir applicability to system requirements.

5.2.3 Sterilization of torpedoes and sea mines. The design of the fuzing system should provide a planned programmed process that renders the weapon permanently incapable of activation after specified events and time when the weapon has served its useful purpose or is no longer capable of functioning.

5.2.4 Explosive ordnance disposal. Fuzes should incorporate Explosive Ordnance Disposal (EOD) feataures based on the applicability of EOD to operational system requirements. (See AFR 136-8)

5.2.5 Inspection and test. The design of the fuze should facilitate the use of inspection and test equipment for visual, physical, or electronic monitoring of all characteristics which assure the safety and intended functioning of the fuze at all appropriate stages. The fuze design should facilitate the use of automatic inspection equipment. Design considerations should include possible connections with the weapon in-flight telemetry test devices.

6. DESIGN GUIDES

6.1 General. Design guides are defined as concepts, logic, background, examples, statements of good practices or rules, time-proven concepts or features, any of which can be used as reminders to aid the thought process of the designer or design agency in an effort to obtain optimum safety through design. The references listed in this section should be used when applicable during the development of fuzes covered by the scope of this document.

6.2 System safety program. Requirements for a system safety program are given in MIL-STD-882. Applicable portions of this standard should be applied throughout the development and life cycle of fuzes.

6.3 Analyses. Analyses are performed to identify hazardous conditions for the purpose of their elimination or control. MIL-STD-882 recommends the performance of Preliminary Hazard Analysis, Subsystem or System Hazard Analysis and Operating Hazard Analysis. A Preliminary Hazard Analysis (pre-design analysis of potential hazards) should be conducted to identify the hazards of abnormal environments, conditions and personnel actions which may occur in the phases before safe separation, and this analysis should be used as a guide to the preparation of design requirements. A system hazard analysis (Failure Modes and Effects Analysis,-Fault Tree Analysis, etc.) should be conducted to evaluate safety of the fuze design, and if quantified, to estimate the safety system failure rate. Techniques for conducting hazard analyses are described in the literature. Examples are given in NAVORD OD 44942 and AFSC Design Handbook DH 1-6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check this source to verify that this is the current version before use.

6.4 Electromagnetic hazards. The general rquirements to preclude hazards resulting from ordnance having electroexplosive devices are provided in MIL-STD-1385. The general requirements and test methods for the design and development of electroexplosive subsystems and associated items to preclude hazards from unintentional initiation are provided in MIL-STD-1512. Applicable portions of these standards should be applied throughout the development and life cycle of fuzes.

7. NOTES

7.1 Suggested color specifications.

7.1.1 Fluorescent Red per TT-L-47; Color No. 38905 per FED-STD-595.

7.1.2 Fluorescent Green per TT-L-47; Color No. 38901 per FED-STD-595.

7.2 Reviewing Activity. Designs which do not meet the requirements of this standard, or which operate on new principles, and for each new application, shall be presented to a review board for a safety evaluation and certification of compliance or appropriate waiver considerations as follows:

7.2.1 For Army: Office of the Product Manager for Fuzes
ATTN: AMCPM-FZ
Picatinny Arsenal, NJ 07806-5000

7.2.2 For Navy: Chairman, Weapon Systems Explosive
Safety Review Board
U.S. Naval Sea Systems Command
Washington, DC 20362

7.2.3 For Air Force: USAF Nonnuclear Munitions Safety Board
ATTN: AD/SES
Eglin Air Force Base, FL 32542

7.3 International agreements. Certain provisions of this standard are the subject of international standardization agreements (ASCC-AIR-STD-20/9, Design Safety Principles for Airborne Weapon Fuzing Systems and STANAG 3525, Design Safety Principles and General Design Criteria for Airborne Weapon Fuzing Systems for Fixed Wing Aircraft and Helicopters). When change notice, revision or cancellation of this document is proposed which affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Supersedes page 12 of MIL-STD-1316C

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

MILITARY STANDARD

FUZE DESIGN, SAFETY CRITERIA FOR

TO ALL HOLDERS OF MIL-STD-1316C:

1. The following pages of MIL-STD-1316C have been revised and supersede the pages listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
11	28 September 1987	11	REPRINTED WITHOUT CHANGE
12	2 November 1987	12	28 September 1987

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-1316C will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

Custodians:
Army - AR
Navy - OS
Air Force - 11

Review Activities:
Army - MI
Navy - AS
Air Force - 18, 99

Preparing Activity:
Army - AR

(Project 1390-0585)

AMSC N/A

FSC 1390

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Approved for public release; distribution is unlimited.
Check the source to verify that this is the current version before use.

5.2.2 Fail-safe features. Fuze systems should incorporate fail-safe features based onltheir applicability to system requirements.

5.2.3 Sterilization of torpedoes and sea mines. The design of the fuzing system should provide a planned programmed process that renders the weapon permanently incapable of activation after specified events and time when the weapon has served its useful purpose or is no longer capable of functioning.

5.2.4 Explosive ordnance disposal. Fuzes should incorporate Explosive Ordnance Disposal (EOD) feataures based on the applicability of EOD to operational system requirements. (See AFR 136-8)

5.2.5 Inspection and test. The design of the fuze should facilitate the use of inspection and test equipment for visual, physical, or electronic monitoring of all characteristics which assure the safety and intended functioning of the fuze at all appropriate stages. The fuze design should facilitate the use of automatic inspection equipment. Design considerations should include possible connections with the weapon in-flight telemetry test devices.

6. DESIGN GUIDES

6.1 General. Design guides are defined as concepts, logic, background, examples, statements of good practices or rules, time-proven concepts or features, any of which can be used as reminders to aid the thought process of the designer or design agency in an effort to obtain optimum safety through design. The references listed in this section should be used when applicable during the development of fuzes covered by the scope of this document.

6.2 System safety program. Requirements for a system safety program are given in MIL-STD-882. Applicable portions of this standard should be applied throughout the development and life cycle of fuzes.

6.3 Analyses. Analyses are performed to identify hazardous conditions for the purpose of their elimination or control. MIL-STD-882 recommends the performance of Preliminary Hazard Analysis, Subsystem or System Hazard Analysis and Operating Hazard Analysis. A Preliminary Hazard Analysis (pre-design analysis of potential hazards) should be conducted to identify the hazards of abnormal environments, conditions and personnel actions which may occur in the phases before safe separation, and this analysis should be used as a guide to the preparation of design requirements. A system hazard analysis (Failure Modes and Effects Analysis, Fault Tree Analysis, etc.) should be conducted to evaluate safety of the fuze design, and if quantified, to estimate the safety system failure rate. Techniques for conducting hazard analyses are described in the literature. Examples are given in NAVORD OD 44942 and AFSC Design Handbook DH 1-6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

6.4 Electromagnetic hazards. The general rquirements to preclude hazards resulting from ordnance having electroexplosive devices are provided in MIL-STD-1385. The general requirements and test methods for the design and development of electroexplosive subsystems and associated items to preclude hazards from unintentional initiation are provided in MIL-STD-1512. Applicable portions of these standards should be applied throughout the development and life cycle of fuzes.

7. NOTES

7.1 Suggested color specifications.

7.1.1 Fluorescent Red per TT-L-47; Color No. 38905 per FED-STD-595.

7.1.2 Fluorescent Green per TT-L-47; Color No. 38901 per FED-STD-595.

7.2 Reviewing Activity. Designs which do not meet the requirements of this standard, or which operate on new principles, and for each new application, shall be presented to a review board for a safety evaluation and certification of compliance or appropriate waiver considerations as follows:

7.2.1 For Army: Chairman, Army Fuze Safety Review Board
U.S. Army Armament Research, Development and Engineering Center
ATTN: AMCPM-FZ
Picatinny Arsenal, NJ 07806-5000

7.2.2 For Navy: Chairman, Weapon Systems Explosive Safety Review Board
U.S. Naval Sea Systems Command
Washington, DC 20362

7.2.3 For Air Force: USAF Nonnuclear Munitions Safety Board
ATTN: AD/SES
Eglin Air Force Base, FL 32542

7.3 International agreements. Certain provisions of this standard are the subject of international standardization agreements (ASCC-AIR-STD-20/9, Design Safety Principles for Airborne Weapon Fuzing Systems and STANAG 3525, Design Safety Principles and General Design Criteria for Airborne Weapon Fuzing Systems for Fixed Wing Aircraft and Helicopters). When change notice, revision or cancellation of this document is proposed which affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Supersedes page 12 of Notice 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph			Page
	1.	SCOPE	1
	1.1	Purpose	1
	1.2	Application	1
	1.3	Excluded munitions	1
	2.	REFERENCED DOCUMENTS	2
	2.1	Issues of documents	2
	2.2	Other publications	3
	3.	DEFINITIONS	4
	4.	REQUIREMENTS	7
	4.1	General	7
	4.2	Fuze Safety System	7
	4.3	Explosive Trains	7
	4.4	Logic Functions	9
	4.5	Safe or Armed Condition	9
	4.6	Safety System Failure Rate	10
	4.7	Documentation	10
	4.8	Design for Quality Control and Inspection	10
	4.9	Design Approval	10
	5.	OBJECTIVES	10
	5.1	General	10
	5.2	Design Features	10
	6.	DESIGN GUIDES	11
	6.1	General	11
	6.2	System Safety Program	11
	6.3	Analyses	11
	6.4	Electromagnetic Hazards	11
	7.	NOTES	12
	7.1	Suggested Color Specifications	12
	7.2	Reviewing Activity	12
	7.3	International Agreements	12

TABLE

Table			
	1.	Approved Explosives for all Services	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Purpose. This standard establishes design and safety criteria for Fuzes and for Safing and Arming (S&A) Devices which are subsystems of fuzes.

1.2 Application. This standard applies to the design of fuzes and S&A devices.

1.3 Excluded munitions. This standard does not apply to fuzes and S&A devices for the following:

1.3.1 Nuclear Weapon Systems and Trainers.

1.3.2 Hand Grenades.

1.3.3 Manually emplaced ordnance items.

1.3.4 Flares and Signals dispensed by hand held devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-339	-	Tetryl
MIL-C-440	-	Compositions A3 and A4
MIL-E-14970	-	Explosive Composition A5
MIL-C-21723	-	Composition CH-6
MIL-I-23659	-	Initiators, Electric, Design and Evaluation of
MIL-P-46464	-	Pellets, Tetryl
MIL-E-81111	-	Explosive, Plastic-Bonded Molding Powder (PBXN-5)

NAVY WEAPON SPECIFICATIONS

WS-4660	-	Dipam Explosive
WS-5003	-	HNS, Explosive
WS-12604	-	Explosive, Plastic-Bonded Molding Powder (PBXN-6)

FEDERAL

TT-L-47	-	Lacquer, Fluorescent Type, Synthetic Base in Pressurized Container

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-444	-	Nomenclature and Definitions in the Ammunition Area
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for
MIL-STD-882	-	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-1385	-	Preclusion of Ordnance Hazards in Electromagnetic Fields, General Requirements for
MIL-STD-1455	-	Dispenser and Sub-Munition, Air Delivered, Safety Design and Safety Qualification Criteria for
MIL-STD-1512	-	Electroexplosive Subsystems, Electrically Initiated, Design Requirements and Test Methods
DOD-STD-1463	-	Evaluation of Munitions to Electromagnetic Fields, Requirements for

FEDERAL

FED-STD-595	-	Color

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AFR 136-8	-	Responsibilities of Explosive Ordnance Disposal
AFSC DH 1-6	-	Design Handbook, System Safety
NAVORD OD 44942	-	Weapon System Safety Guidelines Handbook
ADA-086259 Vol. 4	-	Joint Services Safety and Performance Manual for Qualification of Explosives for Military Use

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 The definitions of MIL-STD-444 generally apply to the ammunition terms used in this standard and the definitions of ADA-086259, Vol. 4 apply to the explosive terms. For interpretation of this standard, the following specific definitions apply:

3.1.1 Arm.

3.1.1.1 For fuzes incorporating explosive train interruption, arm shall be defined as to make a fuze ready for functioning by removal of the safety features, including the explosive train interrupter, or by the alignment of the explosive elements in the explosive train of the fuze.

3.1.1.2 For fuzes not incorporating explosive train interruption but containing minimum functioning energy, arm shall be defined as to make a fuze ready for functioning by removal of the interruption to, or by alignment of, any minimum required energy path in the fuze capable of functioning the initiator.

3.1.1.3 For fuzes not incorporating explosive train interruption or functioning energy interruption, arm shall be defined as to make a fuze ready for functioning by removal of the safety features and the accumulation in the fuze of the minimum energy capable of functioning the initiator.

3.1.2 Arming delay. A delay between launch and fuze arming to obtain a safe separation distance.

3.1.3 Assembled fuze. The completed fuze with all component parts put together; a fuze requiring no added components or parts to prepare it for installation into the ammunition in which it is to function. Assembling the fuze is the process of putting the parts and components together.

3.1.4 Booster and lead explosives. Booster and lead explosives are compounds or formulations such as those explosives listed in Table I of 4.3.1 which are used to transmit and augment the detonation reaction.

3.1.5 Dud. A munition which has failed to function although functioning was intended.

3.1.6 Early burst. A malfunction in which the fuze functions after the arming delay but before the fuze should properly operate.

3.1.7 Enabling. The act of removing or activating one or more safety features which prevent arming, thus permitting arming to occur subsequently.

3.1.8 Environment. The total set of physical conditions to which the fuze may be exposed.

3.1.9 Environmental force. A specific stimulus obtained from the environment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

3.1.10 Explosive train. The detonation or deflagration train (i.e., transfer mechanism) beginning with the first explosive element (e.g., primer, detonator) and terminating in the main charge (e.g., munition functional mechanism, high explosive, pyrotechnic compound, etc.).

3.1.11 Fail-safe feature. A characteristic of a fuze system or part thereof which is designed to result in failure of the ammunition to function when safety feature(s) malfunction.

3.1.12 Firmware. Software which resides in non-volatile computer memory that is not alterable by the computer system during program execution.

3.1.13 Fuze. A physical system designed to sense a target or the result of other prescribed conditions such as time, barometric pressure, command, etc., and initiate a train of fire or detonation in an item of ammunition (warhead, projectile, bomb with an explosive charge, pyrotechnic, chemical or other load). Safing and arming are primary functions performed by a fuze to preclude initiation of the ammunition before the desired position or time. Types of fuzes are usually distinguished by modifying terms as part of the item name to indicate methods of sensing the target or methods of initiating the ammunition (for example: impact fuze, proximity fuze, hydrostatic fuze, time fuze, point detonating fuze, barometric fuze, command fuze, etc.). Fuzes will normally contain explosive components directly within the design package.

3.1.14 Fuze installation. The act of installing or inserting the assembled fuze into the ammunition in which it is to function.

3.1.15 Fuze safety system. The aggregate of devices (environment sensors, launch event sensors, command functioned devices, removable critical items, logic networks as examples, plus the explosive train or initiation interrupter, if applicable) included in the fuze to provide safety and which prevents arming and functioning of the fuze through all environments during transportation, storage, handling, installation and launch until the arming delay has been achieved.

3.1.16 Independent safety feature. Each of the several features is independent if it senses an environment different from that of the other safety features and if the integrity of the safety feature is not affected by the function or malfunction of one or more of the other safety features.

3.1.17 Initiator. The initiator is the device which is capable of directly causing functioning of the fuze explosive train.

3.1.18 Interrupted explosive train. An explosive train with the primary explosives physically separated from the output lead and booster explosives by a barrier to interrupt the explosive path and thus prevent detonation of the lead, booster or main charge until arming occurs.

3.1.19 Launch cycle. The period between the time the ammunition is irreversibly committed to launch and the time it leaves the launcher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

3.1.20 Main charge. The explosive charge which is provided to accomplish the end result in ammunition; e.g., bursting a casing to produce blast and fragmentation, splitting a canister to dispense submissiles or producing other effects for which it may be designed. Main charge explosives are compounds or formulations such as TNT or Composition B, which are used as the final charge in any explosive application. These explosives, because of their relative insensitivity, ordinarily require initiation by a booster explosive.

3.1.21 Minimum functioning energy. The threshold energy level, of whatever type, which will produce the required low probability of functioning the initiator.

3.1.22 Non-interrupted explosive train. An explosive train which has no physical interruption of the explosive elements.

3.1.23 Premature. A malfunction in which the fuze functions before completion of the arming delay.

3.1.24 Primary explosives. Primary explosives are sensitive formulations or compounds such as lead azide or lead styphnate which are used to initiate detonation. They are used in primers or detonators of fuzes and are sensitive to heat, impact, or friction and undergo a rapid reaction upon initiation. These sensitive explosives are separated from the output lead and booster explosive by the interrupter of the fuze.

3.1.25 Safe separation distance. A minimum distance between the delivery system (or launcher) and the launched ammunition beyond which the hazards associated with functioning of that ammunition are acceptable.

3.1.26 Safety and arming device. A mechanism whose primary purpose is to prevent an unintended functioning of the main charge of the ammunition due to the fuze prior to completion of the arming delay and in turn, allow the explosive train of the ammunition to function after arming.

3.1.27 Safety feature. An element or combination of elements which prevents unintentional arming or functioning.

3.1.28 Safety system failure. A failure of the fuze to provide its control over arming and initiation of the explosive train or detonation of the main explosive charge.

3.1.29 Safety system failure rate. The expected ratio of safety system failures for all fuzes for all phases from manufacture of the fuzes to the attainment of the designated points in the fuze life cycle.

3.1.30 Sensor, environmental. A component or series of components designed to detect and respond to a specific environment.

3.1.31 Sterilization. To render the fuze permanently incapable of initiating the explosive system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

3.1.32 Stored energy. A component contains stored energy if the component is capable of delivering energy in addition to the external energy required to initiate its function. Examples of stored energy components are:

a. Batteries

b. Charged firing capacitors

c. Compressed gas

d. Explosive actuators (bellows and dimple motors, etc.)

e. Loaded springs

4. REQUIREMENTS

4.1 General. The features, procedures or controls listed in this section shall be used in the design of fuzes included in the scope of this document.

4.2 Fuze safety system.

4.2.1 Safety redundancy. The safety system of fuzes shall contain at least two independent safety features, each of which shall prevent unintentional arming of the fuze. The forces enabling a minimum of two safety features shall be derived from different environments. An action taken to initiate launch may be considered an environment if the signal generated by the action irreversibly commits the ammunition to complete the launch cycle. Operation of at least one of these safety features shall depend on sensing an environment after first motion in the launch cycle or on sensing a post-launch environment.

4.2.2 Arming delay. A safety feature of the fuze shall provide an arming delay. Arming delay when used as the criterion for safety shall assure that a safe separation distance can be achieved for all defined operational conditions.

4.2.3 Manual arming. An assembled fuze shall not be capable of being armed manually.

4.3 Explosive trains.

4.3.1 Explosive sensitivity (lead and booster explosives). Only explosives subjected to the tests of ADA-086259, Vol. 4, Chapters 3 and 4 and specifically approved by the associated safety evaluation board of 7.2 for each application are permitted in a position leading to the initiation of a high explosive main charge without interruption, when the fuze is in the safe condition. The explosive material used in fuze systems shall not be altered by any means (precipitation, recrystallization, grinding, density changes, etc.) likely to increase its sensitivity beyond that at which the material was approved for service use and at which it is customarily used. Table I contains a list of currently approved explosives for all Services.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

TABLE 1. Approved Explosives for all Services

Explosive	Specification
Comp A3	MIL-C-440
Comp A4	MIL-C-440
Comp A5	MIL-E-14970
Comp CH6	MIL-C-21723
PBXN-5	MIL-E-81111
PBXN-6	WS-12604
DIPAM	WS-4660
HNS Type 1 or Type 2 GR A	WS-5003
*Tetryl	MIL-T-339
*Tetryl Pellets	MIL-P-46464

*No longer manufactured. Not for use in new developments.

4.3.2 Explosive train interruption. When the explosive train contains explosive material other than those allowed by 4.3.1, at least one interrupter (shutter, slider, rotor) shall separate those explosives more sensitive than those allowed by 4.3.1 from the lead and booster until the arming sequence is completed. The interrupter(s) shall be directly locked mechanically in the safe position by at least two independent safety features. The safety features shall not be removed until the arming sequence begins. If the initiating explosive material is housed in the interrupter, a single interrupter locked by the two independent safety features is acceptable. If the sensitive element is positioned such that safety is completely dependent upon the presence of an interrupter, the design shall include positive means to prevent the fuze from being assembled without the properly positioned interrupter. The effectiveness of interruption shall be determined by techniques described in MIL-STD-331, Test 115, including Section 6 or equivalent to Section 6, during design evaluation.

4.3.3 Non-interrupted explosive train control. When the explosive train contains only those explosive materials allowed by 4.3.1, no explosive train interruption is required. One of the following methods of controlling function energy shall be employed to preclude arming before safe separation. Additionally, the fuze design shall include positive means to prevent the fuze from being assembled without its energy control feature(s). The combined probability of having minimum function energy in the fuze, having a failure of the energy control feature(s) and firing the initiator with minimum function energy must be compatible with the specified fuze safety system failure rate (see 4.6).

a. For fuzes containing minimum non-electrical function energy prior to safe separation, at least one energy interrupter directly and mechanically locked in the safe position by at least two independent safety features shall prevent the flow of energy to the initiator.

b. For fuzes containing minimum electrical function energy prior to safe separation, at least two energy interrupters directly and mechanically locked in the safe position, each by an independent safety feature, shall prevent the flow of electrical energy to the initiator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

c. For systems using techniques for accumulating functioning energy from the post-launch environment, the fuze shall not permit any functioning energy to reach the initiator until verification, by the fuze, of a proper launch, and attainment of a safe separation distance. Additionally, any energy of the type required to function the initiator which exists in the fuze prior to safe separation shall be less than the minimum function energy.

4.3.4 Electrical sensitivity. The initiator for an electrically fired non-interrupted explosive train shall meet the characteristics listed for Class B initiators of MIL-I-23659. The initiator shall not be capable of being initiated by any electrical signal at a potential of less than 500 volts. Electromagnetic emission sensitivity and susceptibility of the fuze shall not create a hazard. The requirements of MIL-STD-461 (and DOD-STD-1463 for the Army) shall apply.

4.4 Logic functions. Any logic functions related to safety elements performed by the fuze shall be embedded as firmware or hardware.

4.5 Safe or armed condition.

4.5.1 Safe-arm options. One or more of the following options shall be selected or combined in the fuze design:

a. A feature which assures a positive means of determining the safe condition at the time of fuze installation into the munition; for example by visual observation.

b. A feature which prevents installation of an armed, assembled fuze into the munition.

c. A feature which prevents assembling the fuze in the armed or partially armed condition. This single option may be selected only if the fuze contains at least one safety feature which requires an environmental force for operation that precludes accidental or inadvertent arming from handling prior to launch (also see 4.5.4).

4.5.2 Safe-arm indication for bomb fuzes. A bomb fuze must possess a feature(s) which assures a positive means of determining the safe condition of the fuze before and after installation into the bomb.

4.5.3 Visual indication. If visual indication of the safe-armed condition is to be employed in the fuze, visible indicators shall be color coded (see 7.1) to show the fuze condition as follows:

a. Safe condition. Fluorescent green background with the letter S or word SAFE superimposed thereon in white. Colors shall be nonspecular.

b. Armed condition. Fluorescent red background with the letter A or the word ARMED superimposed thereon in black. Colors shall be nonspecular.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

4.5.4 Arming and reset. If arming and reset of the assembled fuze in tests is a normal procedure in manufacturing, inspection, or at any time prior to fuze installation, option 4.5.1 c is not sufficient and the provisions of 4.5.1 a or 4.5.1 b shall be met.

4.6 Safety system failure rate. The safety system failure rate of a fuze shall be calculated by performing a safety analysis (see 6.3) and shall be verified to the extent practicable by test and analysis during fuze evaluation. As a minimum requirement, the safety system failure rate shall not exceed one failure in one million prior to intentional initiation of the arming sequence. A minimum quantitative requirement is not specified for the time frame after launch to the attainment of the proper arming delay since it will be dependent upon the particular weapon and its use. Fuze malfunctions which occur after the proper arming delay shall not be included in the fuze safety system failure rate.

4.7 Documentation. The evaluation program used as the basis of the safety assessment which is prepared by the developing agency shall be documented in both detail and summary form.

4.8 Design for quality control and inspection.

4.8.1 Fuzes shall be designed and documented to facilitate application of effective quality control and inspection procedures. Design characteristics critical to fuze safety shall be identified to assure that the designed safety is maintained.

4.8.2 The design of the fuze shall facilitate the use of inspection and test equipment for all critical design characteristics identified.

4.9 Design approval. At the inception of engineering development, the weapon system program manager shall obtain approval of the cognizant service authority for weapon safety (and other service authorities, for multi-service application) of both the design concept and the methodology for assuring a comprehensive safety assessment. At the completion of engineering development, the adequacy of the safety assessment shall require the review and approval of the cognizant service weapon authority(ies), (see 7.2).

5. OBJECTIVES

5.1 General. The features, procedures, or controls listed as objectives in this section are to be accepted as requirements in the design of fuzes covered by the scope of this document when achievement is within the state of the art of fuze development and is feasible within the scope of the development. Necessity for deviations from these objectives shall be documented, showing that compliance is not feasible (see 7.2).

5.2 Design features.

5.2.1 Stored energy. The fuze safety system should not utilize stored energy to remove safety features or provide arming energy unless no adequate environmentally derived energy is available. If stored energy is used, it must be demonstrated that safety is not compromised.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

5.2.2 Fail-safe features. Fuze systems should incorporate fail-safe features based on their applicability to system requirements.

5.2.3 Sterilization of torpedoes and sea mines. The design of the fuzing system should provide a planned programmed process that renders the weapon permanently incapable of activation after specified events and time when the weapon has served its useful purpose or is no longer capable of functioning.

5.2.4 Explosive ordnance disposal. Fuzes should incorporate Explosive Ordnance Disposal (EOD) features based on the applicability of EOD to operational system requirements. (See AFR 136-8)

5.2.5 Inspection and test. The design of the fuze should facilitate the use of inspection and test equipment for visual, physical, or electronic monitoring of all characteristics which assure the safety and intended functioning of the fuze at all appropriate stages. The fuze design should facilitate the use of automatic inspection equipment. Design considerations should include possible connections with the weapon in-flight telemetry test devices.

6. DESIGN GUIDES

6.1 General. Design guides are defined as concepts, logic, background, examples, statements of good practices or rules, time-proven concepts or features, any of which can be used as reminders to aid the thought process of the designer or design agency in an effort to obtain optimum safety through design. The references listed in this section should be used when applicable during the development of fuzes covered by the scope of this document.

6.2 System safety program. Requirements for a system safety program are given in MIL-STD-882. Applicable portions of this standard should be applied throughout the development and life cycle of fuzes.

6.3 Analyses. Analyses are performed to identify hazardous conditions for the purpose of their elimination or control. MIL-STD-882 recommends the performance of Preliminary Hazard Analysis, Subsystem or System Hazard Analysis and Operating Hazard Analysis. A Preliminary Hazard Analysis (pre-design analysis of potential hazards) should be conducted to identify the hazards of abnormal environments, conditions and personnel actions which may occur in the phases before safe separation, and this analysis should be used as a guide to the preparation of design requirements. A system hazard analysis (Failure Modes and Effects Analysis, Fault Tree Analysis, etc.) should be conducted to evaluate safety of the fuze design, and if quantified, to estimate the safety system failure rate. Techniques for conducting hazard analyses are described in the literature. Examples are given in NAVORD OD 44942 and AFSC Design Handbook DH 1-6.

6.4 Electromagnetic hazards. The general requirements to preclude hazards resulting from ordnance having electroexplosive devices are provided in MIL-STD-1385. The general requirements and test methods for the design

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

and development of electroexplosive subsystems and associated items to preclude hazards from unintentional initiation are provided in MIL-STD-1512. Applicable portions of these standards should be applied throughout the development and life cycle of fuzes.

7. NOTES

7.1 Suggested color specifications.

7.1.1 Fluorescent Red per TT-L-47; Color No. 38905 per FED-STD-595.

7.1.2 Fluorescent Green per TT-L-47; Color No. 38901 per FED-STD-595.

7.2 Reviewing Activity. Designs which do not meet the requirements of this standard, or which operate on new principles, and for each new application, shall be presented to a review board for a safety evaluation and certification of compliance or appropriate waiver considerations as follows.

7.2.1 For Army: Commander
Armament Research & Development Center
Army Fuze Management Office
ATTN: DRSMC-ASF(D)
Dover, NJ 07801

7.2.2 For Navy: Chairman, Weapon Systems Explosive
Safety Review Board
US Naval Sea Systems Command
Washington, DC 20362

7.2.3 For Air Force: USAF Nonnuclear Munitions Safety Board
ATTN: AD/SES
Eglin Air Force Base, FL 32542

7.3 International agreements. Certain provisions of this standard are the subject of international standardization agreements (ASCC-AIR-STD-20/9, Design Safety Principles for Airborne Weapon Fuzing Systems and STANAG 3525, Design Safety Principles and General Design Criteria for Airborne Weapon Fuzing Systems for Fixed Wing Aircraft and Helicopters). When change notice, revision or cancellation of this document is proposed which affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

Custodians:

Army-AR
Navy-OS
Air Force-11

Review Activities:

Army-AR, MI
Navy-OS, AS
Air Force-11, 18, 99

Preparing Activity:

Army-AR

Project Number:

1390-0334

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:41Z
Check the source to verify that this is the current version before use.